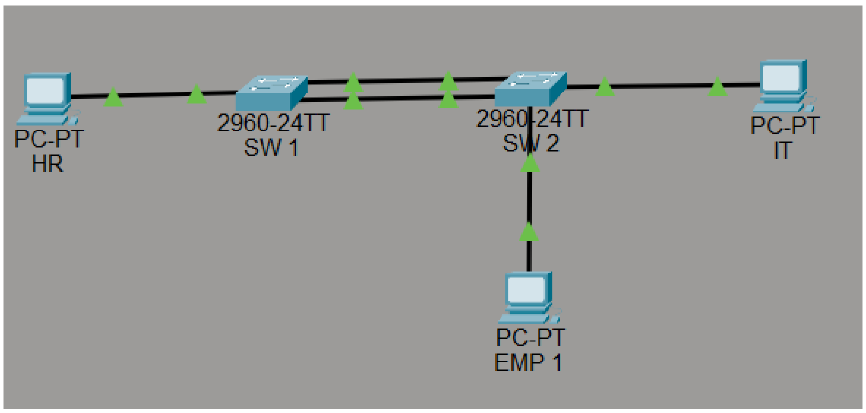
MINI PROJECT 6

EtherChannel using PAgP + Trunking + STP Interaction (Cisco-Proprietary)

DIAGRAM:



DESCRIPTION:

This project explains how EtherChannel using PAgP combines multiple physical links into one logical link for higher bandwidth and redundancy.

STEP 1: Create VLANs on SW 1,SW2:

Verify existing VLANs from Previous projects

STEP 2: Assign Access Ports to VLANs on SW1 & SW2:

SW1:

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 10
exit
```

SW 2:

interface fastEthernet0/1
switchport mode access
switchport access vlan 20
exit
interface fa0/2
switchport mode access
switchport access vlan 10
exit

STEP 3: Configure PAgP EtherChannel

Sw1,
interface range fastEthernet0/23 - 24
channel-group 1 mode **desirable**
exit
sw2,
interface range fastEthernet0/23 - 24
channel-group 1 mode **auto**
exit
passive on both sides will not create etherchannel

STEP 4 : Configure Port-Channel as Trunk:

interface port-channel 1
switchport mode trunk
switchport trunk allowed vlan 10,20
exit

STEP 5: Configure IP Addressing on PCs:

PC	VLAN	IP Address	Gateway
PC-HR	10	192.168.10.10/24	192.168.10.1
PC-IT	20	192.168.20.10/24	192.168.20.1
PC-EMP 1	10	192.168.10.11/24	192.168.10.1

STEP 6: Verification & STP Observation:

show etherchannel summary

```
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           PAgP   Fa0/23(P) Fa0/24(P)
Switch#
Switch#
```

show spanning-tree

<u>STEP 7: Ping Test:</u>

```
HR
Physical  Config  Desktop  Programming  Attributes
Command Prompt                                                    X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time=2ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

we do not have Router or Layer 3 switch, so ping test from HR-PC to EMP 1.